

Project Report: Enterprise Network Deployment

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Technical Focus: Network Infrastructure / Cybersecurity / System Administration

Contents

Lab Project #1: The "Connectivity Baseline"	3
Lab Project #2: The "Inter-Network Bridge"	5
Lab Project #3: The "Virtual Terminal"	8
Lab Project #4: The "Secure Shell" (SSH)	11
Lab Project #5: The "Hardened Gateway"	15

Lab Project #1: The "Connectivity Baseline"

Goal: Build a basic network from scratch and verify that Layer 3 (IP) is working correctly.

Task:

1. Open Cisco Packet Tracer.
2. Place two PCs and one 2960 Switch.
3. Connect both PCs to the switch using Copper Straight-Through cables
4. Configure the IP addresses manually:

Addressing Table:

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC-0	FastEthernet0	192.168.1.10	255.255.255.0	N/A
PC-1	FastEthernet0	192.168.1.11	255.255.255.0	N/A
Switch-0	FastEthernet0/1 FastEthernet0/2	N/A	N/A	N/A

5. Wait for the lights on the cables to turn green.
6. Open the Command Prompt on PC-0 and try to ping PC-1.

Status Report:

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

Figure 1: Successful ICMP Echo Request/Reply

- The above image (figure 1) confirms that the ping is successful. This verifies that **ICMP (Internet Control Message Protocol)** is successfully communicating via **Unicast** traffic across the switch and that the **NICs (Network Interface Cards)** are correctly configured with the proper IP/Subnet mask.

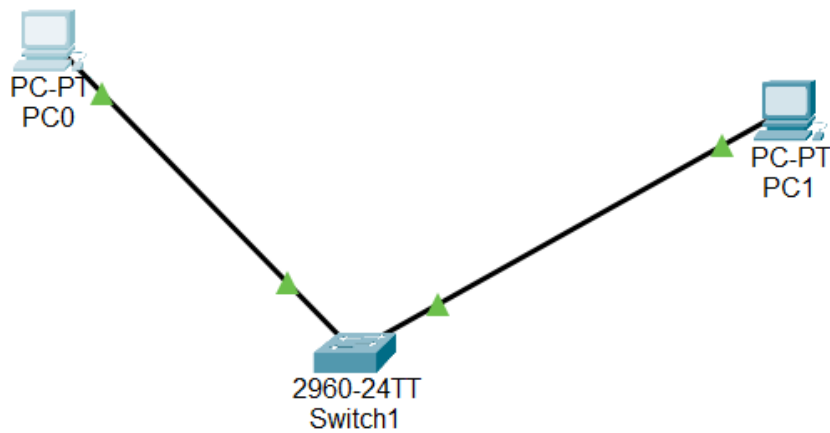


Figure 2: Lab 1 Physical Topology

- The above image (figure 2) shows the final network topology: a simple Peer-to-Peer **Local Area Network (LAN)** connected via a **Layer 2 Catalyst 2960 Switch**. The green link lights indicate that the Physical Layer (Layer 1) and Data Link Layer (Layer 2) are fully operational.

Lab Project #2: The "Inter-Network Bridge"

Goal: Use a Router to connect two different networks so they can communicate.

Task:

1. Open Cisco Packet Tracer.
2. Place two PCs, two 2960 Switches, and one 4331 Router.
3. Connect the hardware using Copper Straight-Through cables:
 - PC-0 ↔ Switch-0
 - Switch-0 ↔ Router
 - PC-1 ↔ Switch-1
 - Switch-1 ↔ Router
4. Configure the IP addresses manually:

Addressing Table:

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC-0	FastEthernet0	192.168.10.10	255.255.255.0	192.168.10.1
PC-1	FastEthernet0	192.168.20.10	255.255.255.0	192.168.20.1
Switch-0	FastEthernet0/1 FastEthernet0/2	N/A	N/A	N/A
Switch-1	FastEthernet0/1 FastEthernet0/2	N/A	N/A	N/A
Router-0	Gig0/0/0 (GigabitEthernet)	192.168.10.1	255.255.255.0	N/A
Router-0	Gig0/0/1 (GigabitEthernet)	192.168.20.1	255.255.255.0	N/A

5. Wait for the lights on the cables to turn green.

Unlike switches, router ports are turned off by default. To turn them on and get the green light, enter these simple commands in the Router CLI tab:

For the Left Side (Gig0/0/0):

```
enable
configure terminal
```

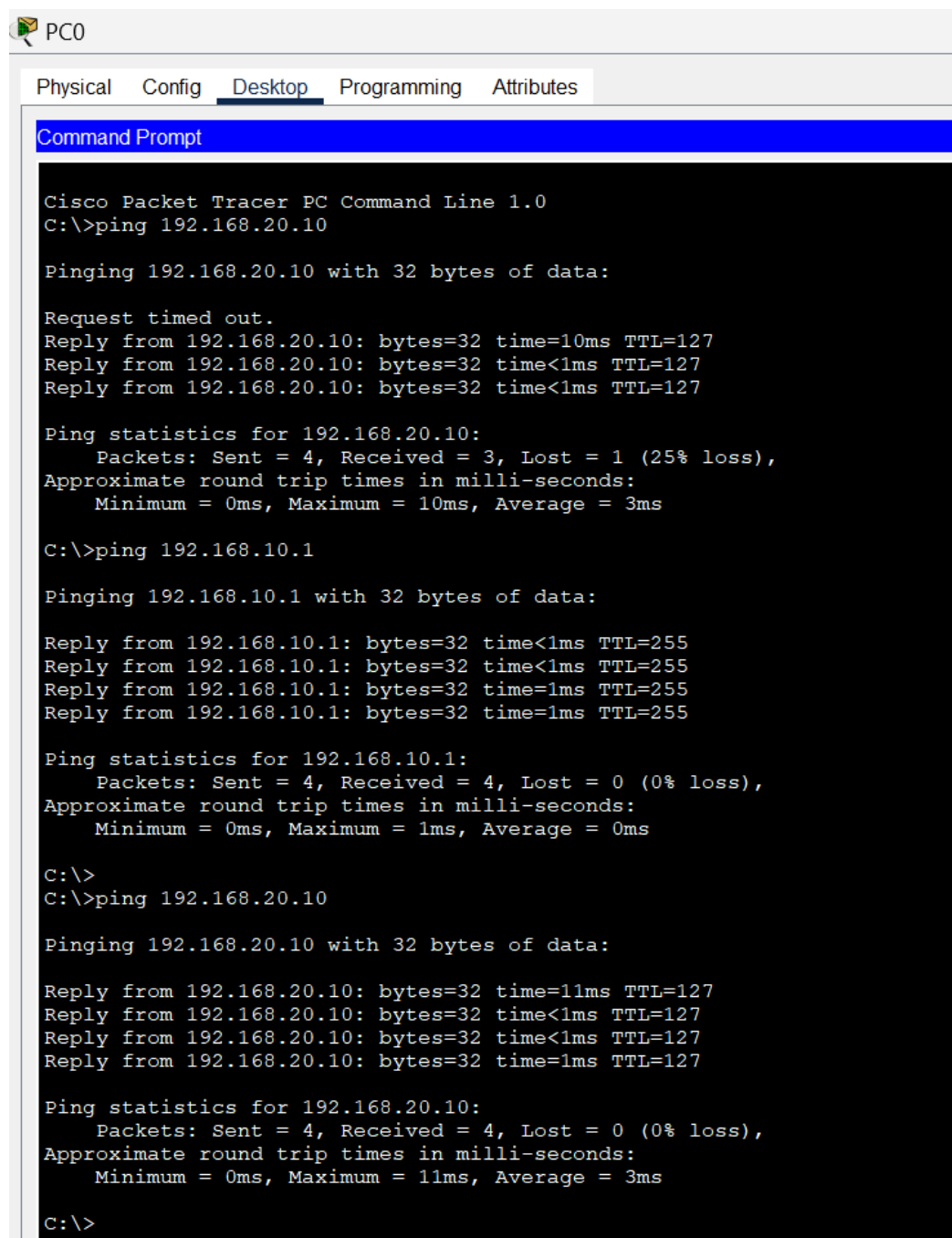
```
interface gigabitEthernet 0/0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
```

For the Right Side (Gig0/0/1):

```
interface gigabitEthernet 0/0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
```

6. Open the Command Prompt on PC-0 and try to ping PC-1.

Status Report:



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time=10ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time=11ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 3ms

C:\>
```

Figure 3: Successful ICMP Echo Request/Reply across remote subnets via Router-0

- The above image (figure 3) confirms that the ping is successful. **ICMP (Internet Control Message Protocol)** is successfully communicating across a routed boundary between the 192.168.10.0 and 192.168.20.0 networks.
- As seen in the image above (figure 3), the first ping request timed out. This is a normal behaviour of the **ARP (Address Resolution Protocol)** process; the router must broadcast an ARP request to map the IP address of the destination to its MAC address before the first ICMP packet can be encapsulated and sent.

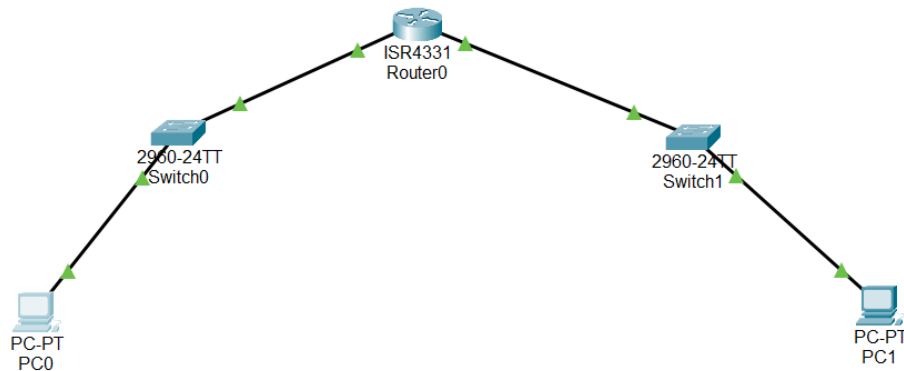


Figure 4: Inter-Network Topology featuring segmented LANs connected by a Cisco 4331 Router

- The above image (figure 4) shows the final network topology. This topology demonstrates a classic **Gateway** configuration. By assigning the Router's interface IPs (192.168.10.1 and 192.168.20.1) as the **Default Gateways** for the PCs, traffic can now move beyond the local switch.

Lab Project #3: The "Virtual Terminal"

Goal: Configure a switch so that it can be managed remotely using Telnet, allowing you to access the Command Line Interface (CLI) from a PC without needing a physical console cable.

Task:

1. Open Cisco Packet Tracer.
2. Place one PC and one 2960 Switch.
3. Connect the PC to the switch using a Copper Straight-Through cable.
4. Configure the IP addresses manually according to the table below.

Addressing Table:

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC-0	FastEthernet0	192.168.1.10	255.255.255.0	N/A
Switch-0	FastEthernet0/1 (VLAN 1)	192.168.1.2	255.255.255.0	N/A

5. On the Switch (CLI Tab):

Unlike a PC, a switch doesn't have a "Desktop" tab for IP configuration. You must assign the IP to its management interface (VLAN 1) and set up "Virtual Terminal" lines for remote access. Click the Switch, go to the CLI tab, and press Enter.

Type the following commands:

```
enable
configure terminal
interface vlan 1
ip address 192.168.1.2 255.255.255.0
no shutdown
exit
```

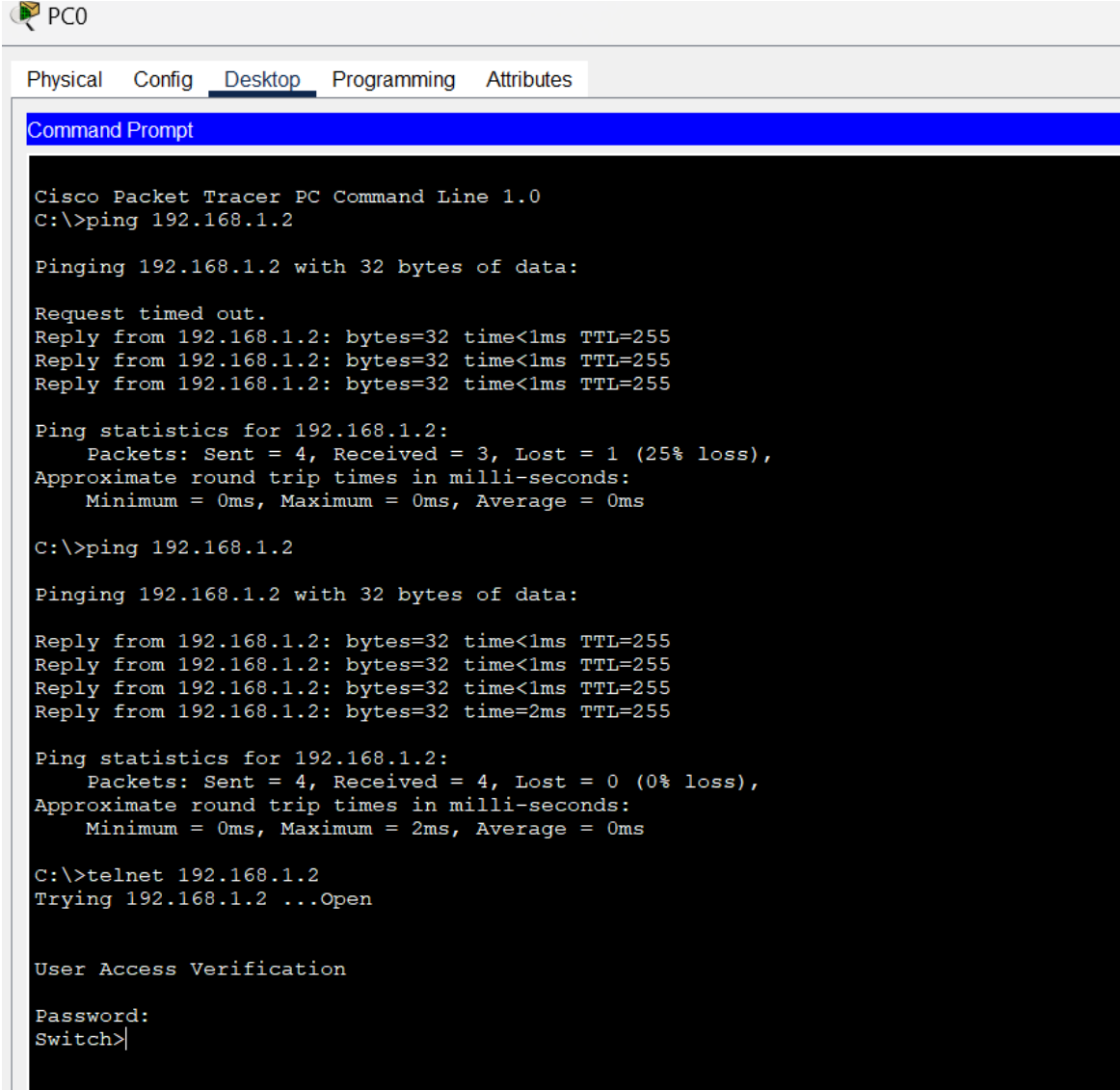
```
# Setting up the Telnet Password
line vty 0 15
password cisco
login
exit
```

```
# Setting the Secret Password (required for remote access to Enable mode)
enable password cisco
```

6. Open the Command Prompt on PC-0.
Type ping 192.168.1.2, to ensure you have a connection.

7. Once the ping is successful, type `telnet 192.168.1.2`.
8. When asked for a password, type `cisco` (Note: the characters won't show as you type).
9. If you see the `Switch>` prompt on your PC screen, you are officially managing the network remotely!

Status Report:



The screenshot shows a PC window titled 'PC0' with tabs for Physical, Config, Desktop, Programming, and Attributes. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The Command Prompt shows the output of a Telnet session to 192.168.1.2. It starts with a ping command, which fails initially with a 'Request timed out.' message. After a second ping, it shows successful results: 4 packets sent, 4 received, 0% loss, and an average round trip time of 0ms. Then, the user enters the telnet command, and the connection is established, showing 'Trying 192.168.1.2 ...Open' and 'User Access Verification'. The prompt changes to 'Password:', and the user enters 'cisco', resulting in the 'Switch>' prompt.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time=2ms TTL=255

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>telnet 192.168.1.2
Trying 192.168.1.2 ...Open

User Access Verification

Password:
Switch>
```

Figure 5: Successful Remote Management Session via Telnet.

- The above image (figure 5) confirms successful access to the Switch CLI from PC-0. This verifies that the **SVI (Switch Virtual Interface)** on VLAN 1 is active and that the **VTY (Virtual Teletype)** lines are correctly accepting connections.
- During the Telnet login, the password characters do not appear on the screen (the cursor remains stationary). This prevents "shoulder surfing" (people nearby seeing the password).
- However, Telnet does not use encryption. Even though the password is hidden on my screen, it is sent across the network as **Plain Text** (for this reason, in the real world, SSH is used-coming up in the next lab project). If a malicious actor were "sniffing"

the network traffic, they could capture the packets and see the password `cisco` in clear text.



Figure 6: Management Plane Topology—PC-0 acting as a Remote Administration Station.

- The above image (figure 6) shows a point-to-point connection between a management PC and a Catalyst 2960 Switch. By assigning an IP to **VLAN 1**, we have created a management "door" into the switch that can be accessed from any authorized device on the subnet.

Lab Project #4: The "Secure Shell" (SSH)

Goal: Replace the insecure Telnet access from Lab Project #3 with SSH, requiring a specific username and encryption key to protect the management session.

Task:

1. Open Cisco Packet Tracer.
2. Place one PC and one 2960 Switch.
3. Connect the PC to the switch using a Copper Straight-Through cable.
4. Configure the IP addresses manually according to the table below.

Addressing Table:

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC-0	FastEthernet0	192.168.1.10	255.255.255.0	N/A
Switch-0	FastEthernet0/1 (VLAN 1)	192.168.1.2	255.255.255.0	N/A

5. On the Switch (CLI Tab):

To enable SSH, the switch needs a "personality" (Hostname and Domain Name) so it can generate a unique digital lock (RSA Key).

- a. Click the Switch, go to the CLI tab, and press Enter.
- b. Unlike a PC, a switch doesn't have a "Desktop" tab for IP configuration. You must assign the IP to its management interface (VLAN 1) and set up "Virtual Terminal" lines for remote access.

Click the Switch, go to the CLI tab, and press Enter.

Enter the following commands:

```
enable
configure terminal
interface vlan 1
ip address 192.168.1.2 255.255.255.0
no shutdown
exit
```

- c. Enter the following commands to set the identity and generate keys:

```
enable
configure terminal
hostname Project4-SW
ip domain-name cisco.com
```

```
# Generating the Encryption Key
crypto key generate rsa
# (When prompted for bits, type 1024 and press Enter)
```

d. Create a local user and lock the virtual doors to only allow SSH:

```
username admin password cisco
line vty 0 15
login local
transport input ssh
exit

# Set secret password for enable mode
enable password cisco
```

6. Open the PC-0 Desktop:

- Instead of the Command Prompt, find the Telnet/SSH Client tool.
- Enter the Host IP: 192.168.1.2
- Enter the Username: admin
- Click Connect and enter the password: cisco
- If you see the `Project4-SW>` prompt, you are now managing the switch through an encrypted "tunnel."

Status Report:

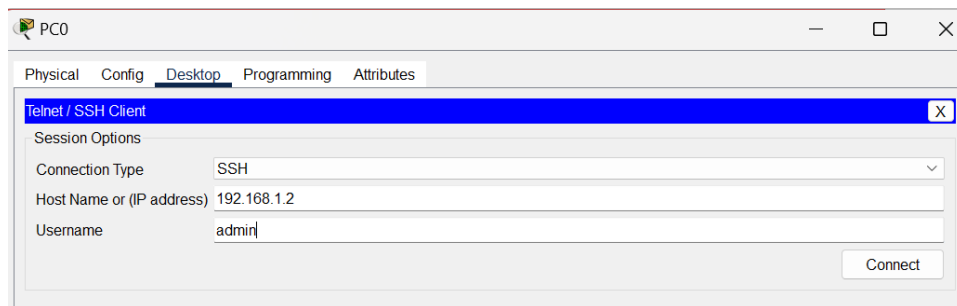


Figure 7: Successful SSH Management Session

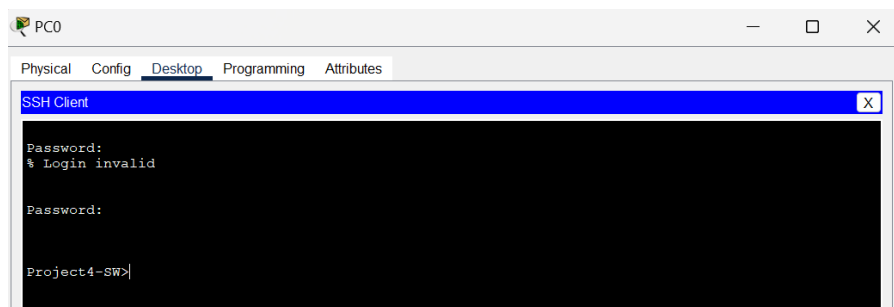


Figure 8: Successful SSH Management Session

- The above images (figures 7 & 8) confirms successful access to the Switch CLI via the **Telnet/SSH Client** tool on PC-0. Unlike Telnet, this connection requires a local **username** (admin) and **password** for authentication.

- This verifies that the **RSA encryption keys** (1024-bit) were correctly generated and that the **VTY lines** are now utilizing the `login local` database for enhanced security.

```
C:\>ssh -l admin 192.168.1.2

Password:

Project4-SW>exit

[Connection to 192.168.1.2 closed by foreign host]
C:\>
C:\>
```

Figure 9: Initiating Secure Management via CLI

- This image (figure 9) shows the transition from a simple connectivity test (ping) to an active secure session. By using the command prompt instead of a GUI tool, I am manually specifying the **Login User** (admin) and the **Target IP** (192.168.1.2).
- This demonstrates the ability to manage network infrastructure using the native command-line SSH client. This is the standard method used by NetSec professionals to access remote equipment. It confirms that the switch's SSH service is listening and ready to challenge the user for their encrypted credentials.

```
C:\>
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>telnet 192.168.1.2
Trying 192.168.1.2 ...Open

[Connection to 192.168.1.2 closed by foreign host]
C:\>
```

Figure 10: Verification of Blocked Insecure Protocols

- This image (figure 10) shows the failed attempt to connect via Telnet. By configuring transport input ssh, we have effectively "locked the door" on unencrypted management traffic.
- **Cisco Pro-Tip:** SSH operates on **Port 22**, whereas Telnet uses **Port 23**. Because SSH encrypts the entire session, any attacker "sniffing" the network will only see scrambled gibberish instead of administrative credentials.

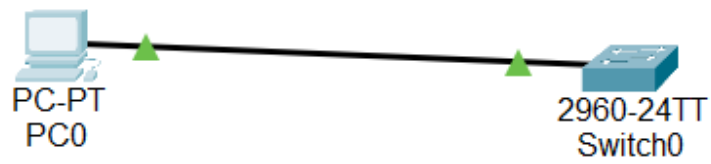


Figure 11: Lab 4 Physical Topology

- The final topology (figure 11) for this lab. While physically identical to Lab 3, the logical management plane is now secured through an **encrypted "tunnel"**.

Lab Project #5: The "Hardened Gateway"

Goal: Build a secure, routed network and implement a "Secure Management Only" policy that blocks insecure protocols across the entire infrastructure.

Task:

1. Open Cisco Packet Tracer.
2. Place the hardware: Two PCs, two 2960 Switches, and one 4331 Router.
3. Connect the hardware: Use Copper Straight-Through cables to connect the PCs to the switches and the switches to the router interfaces specified below.
4. Configure IP addresses: Manually assign the addresses according to this table:

Addressing Table:

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC-0	FastEthernet0	192.168.10.10	255.255.255.0	192.168.10.1
PC-1	FastEthernet0	192.168.20.10	255.255.255.0	192.168.20.1
Switch-0	FastEthernet0/1 FastEthernet0/2 (VLAN 1)	192.168.10.2	255.255.255.0	192.168.10.1
Switch-1	FastEthernet0/1 FastEthernet0/2 (VLAN 1)	192.168.20.2	255.255.255.0	192.168.20.1
Router-0	Gig0/0/0 (GigabitEthernet)	192.168.10.1	255.255.255.0	N/A
Router-0	Gig0/0/1 (GigabitEthernet)	192.168.20.1	255.255.255.0	N/A

Unlike a PC, a switch doesn't have a "Desktop" tab for IP configuration. You must assign the IP to its management interface (VLAN 1) and set up "Virtual Terminal" lines for remote access.

Click the Switch, go to the CLI tab, and press Enter.

Enter the following commands:

Bash

```
enable
configure terminal
interface vlan 1
ip address 192.168.10.2 255.255.255.0
no shutdown
exit
ip default-gateway 192.168.10.1
```

(Repeat similarly for Switch-1 using its IP 192.168.20.2 and Gateway 192.168.20.1)

5. Enable Routing & Management

On the Router CLI, enable the ports and set up the management "personality":

Bash

```
Router> enable
Router# configure terminal
Router(config)# hostname Gateway-Router
Router(config)# ip domain-name cisco.com
Router(config)# interface gigabitEthernet 0/0/0
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface gigabitEthernet 0/0/1
Router(config-if)# ip address 192.168.20.1 255.255.255.0
Router(config-if)# no shutdown
```

6. The "Hardening" (Security Layer)

Now, lock down the router so it only allows secure management via SSH.

a. Generate the Encryption Key:**Bash**

```
Gateway-Router(config)# crypto key generate rsa
# (Type 1024 and press Enter)
```

b. Create the Admin and Lock the Doors:**Bash**

```
Gateway-Router(config)# username admin password cisco
Gateway-Router(config)# enable password cisco
Gateway-Router(config)# line vty 0 4
Gateway-Router(config-line)# login local
Gateway-Router(config-line)# transport input ssh
# (This blocks Telnet completely!)
Gateway-Router(config-line)# exit
```

```

Gateway-Router(config)# service password-encryption
# (This ensures that all local passwords in the configuration are now
encrypted and not visible in plain text).
Gateway-Router(config)# end

```

```

Gateway-Router# show running-config

```

```

Gateway-Router(config)#service password-encryption
Gateway-Router(config)#end
Gateway-Router#
%SYS-5-CONFIG_I: Configured from console by console

Gateway-Router#show running-config
Building configuration...

Current configuration : 803 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Gateway-Router
!
!
!
enable password 7 0822455D0A16
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!

Gateway-Router#
Gateway-Router#

```

Figure 12: Verification of Global Password Encryption

The above screenshot (figure 12) verifies the successful application of the “service password-encryption” command. As shown, the local passwords are now "scrambled" (Type 7 encryption) rather than appearing in plain text. This is a critical hardening step to prevent credential exposure from configuration files.

```

Gateway-Router# copy running-config startup-config
# (Final save to make all changes permanent)
Gateway-Router# exit

```

```

Gateway-Router>enable
Password:
Gateway-Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Gateway-Router#exit

```

Figure 13: Administrative Persistence (NVRAM Save)

The above screenshot (figure 13) shows a confirmation of committing the current configuration to the **startup-config** (NVRAM). This ensures that all security hardening measures and IP configurations persist even if the router is rebooted or loses power.

Status Report:

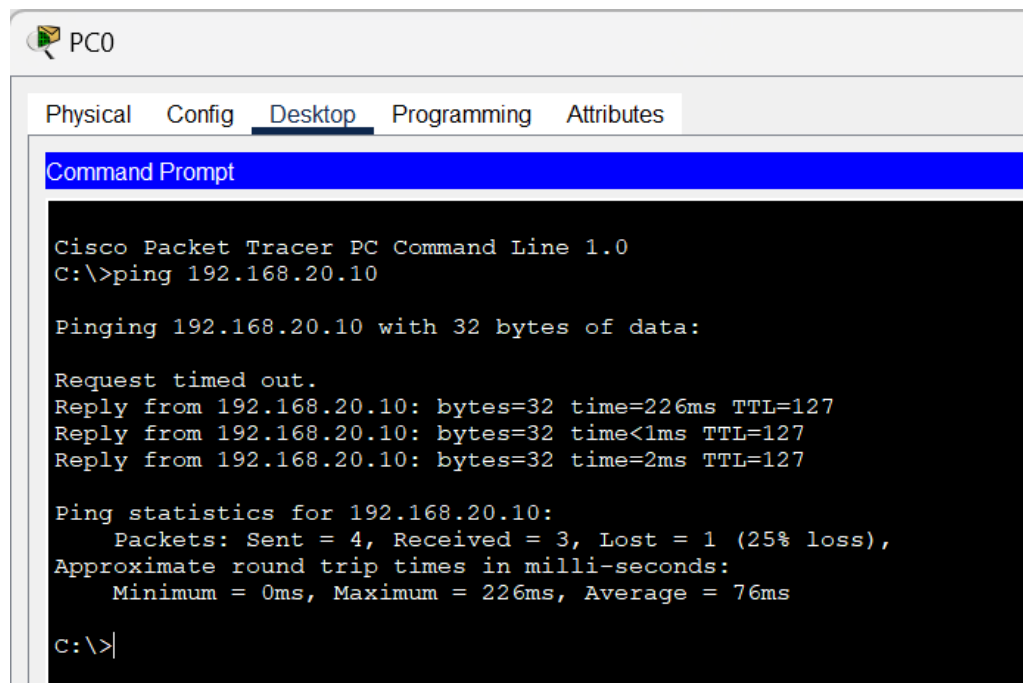


Figure 14: Inter-Network Connectivity Check (Layer 3)

The above image (figure 14) confirms that **Inter-VLAN routing** is functional. Despite PC-0 and PC-1 being on different subnets (10.0 and 20.0), the gateway-router successfully routed the ICMP traffic between the two interfaces.

```

C:\>telnet 192.168.10.1
Trying 192.168.10.1 ...Open

[Connection to 192.168.10.1 closed by foreign host]
C:\>

```

Figure 15: Lockdown Verification (Telnet Blocked)

The above image (figure 15) verifies that the security policy is being enforced. While the ping (ICMP) was successful, the attempt to connect via Telnet was rejected by the router. This proves the “transport input ssh” command is successfully protecting the management plane from insecure protocols.

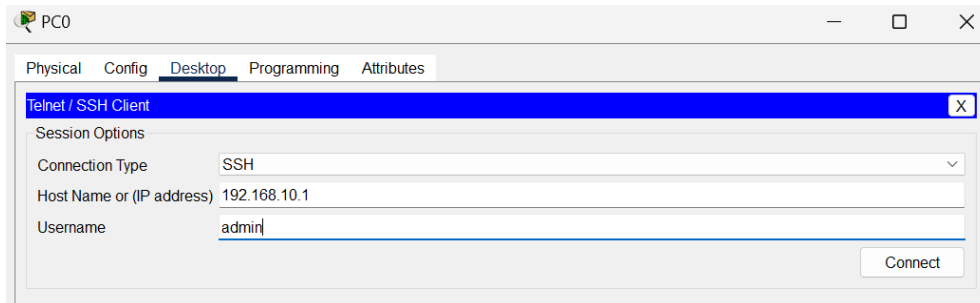


Figure 16: Secure Remote Management Initialization

The above image (figure 16) shows the initialization of a secure session. By explicitly choosing **SSH** and providing the **admin** username, we are adhering to the principle of "Least Privilege" and encrypted access.

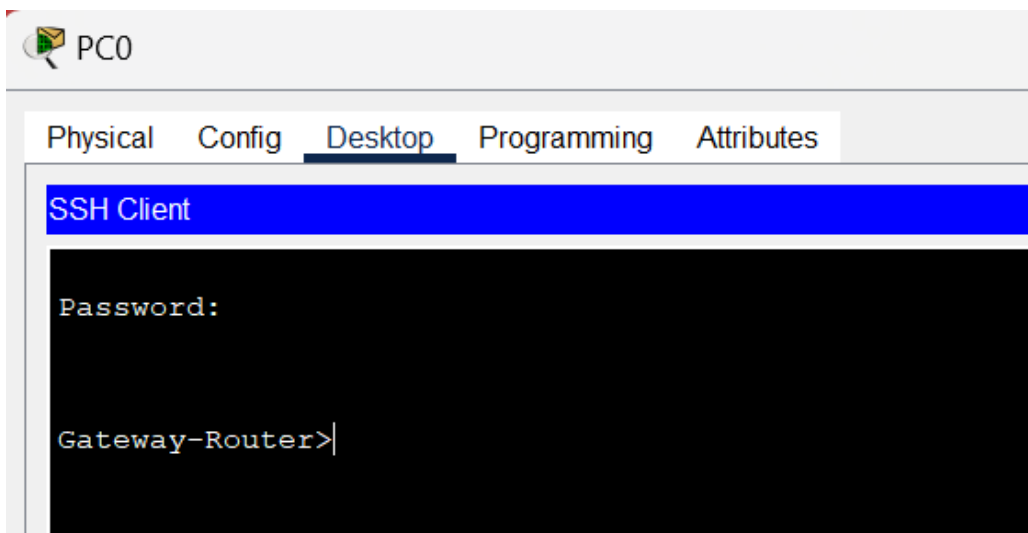


Figure 17: Successful SSH Management Session

The above image (figure 17) confirms a fully encrypted remote session to the **gateway-router**. This demonstrates that the router is now part of a secure management infrastructure, accessible only through encrypted channels.

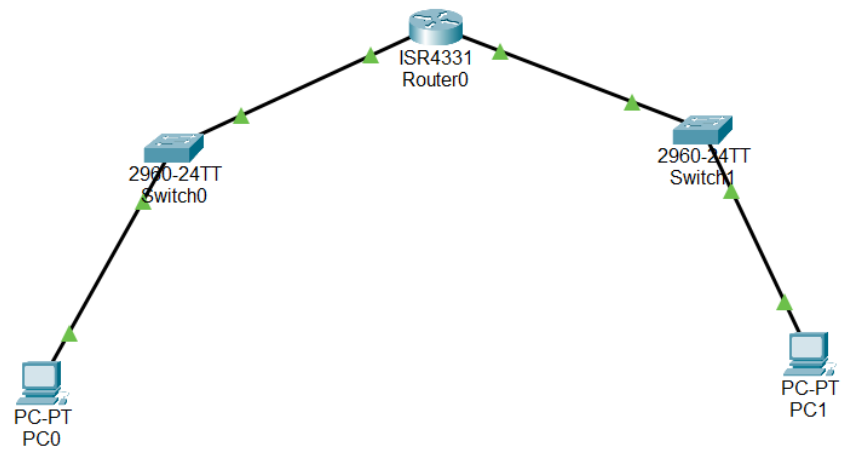


Figure 18: Final Hardened Enterprise Topology

The above image (figure 18) shows the final physical and logical layout of the "Hardened Gateway" project. This architecture features redundant subnets, a centralized gateway, and a fully secured management plane.